www.tntextbooks.in

of carpals with the ligaments. This tunnel is termed as **carpal tunnel**.

Metacarpals,the palm bones are 5 in number and **phalanges** the digits bones are 14 in number.

(b) Pelvic Girdle

The pelvic girdle is a heavy structure specialized for weight bearing. It is composed of two hip bones called coxal bones that secure the lower limbs to the axial skeleton (Figure 9.11). Together, with the sacrum and coccyx, the hip bones form the basin-like bony pelvis.

Each coxal bone consists of three fused bones, **ilium, ischium** and **pubis**. At the point of fusion of ilium, ischium, and pubis a deep hemispherical socket called the acetabulum is present on the lateral surface of the pelvis. It receives the head of the femur or thigh bone at the hip joint and helps in the articulation of the femur. Ventrally the two halves of the pelvic girdle meet and form the **pubic symphysis** containing fibrous cartilage.

The **ilium** is the superior flaring portion of the hip bone. Each ilium forms a secure joint with the sacrum posteriorly. The **ischium** is a curved bar of bone. The V-shaped **pubic bones** articulate anteriorly at the **pubic symphysis**. The pelvis of male is deep and narrow with larger heavier bones and the female is shallow, wide and flexible in nature, and this helps during pregnancy which is influenced by female hormones.

The Lower limb

The lower limb consists of 30 bones which carries the entire weight of the erect body

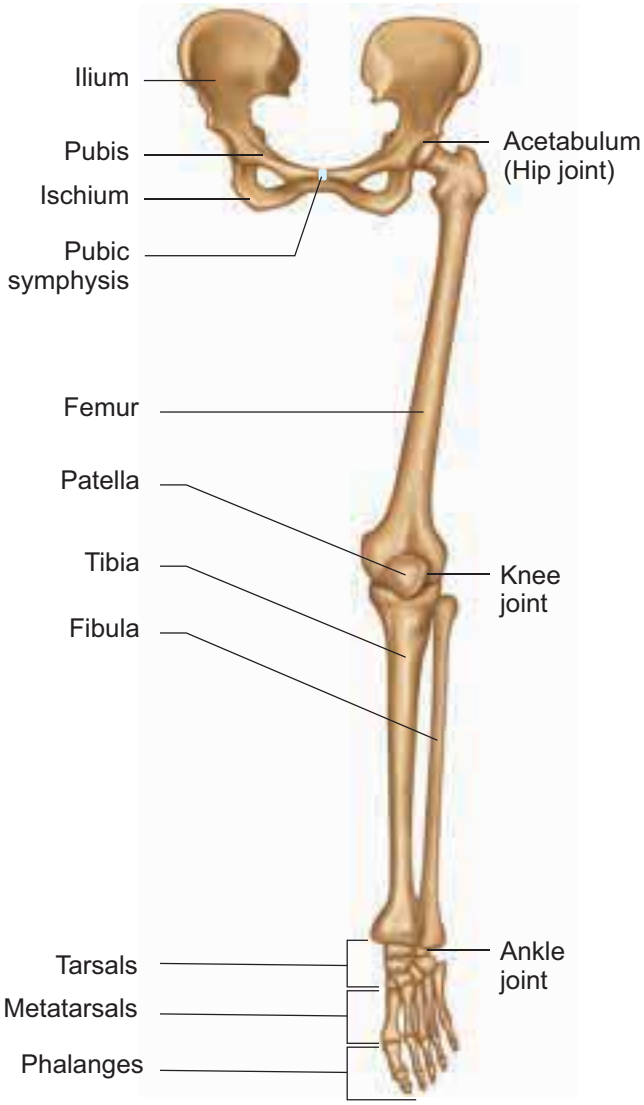

Figure 9.11 Pelvic girdle with lower limb

and is subjected to exceptional forces when we jump or run. The bones of the lower limbs are thicker and stronger than the upper limbs. The three segments of each lower limb are **the thigh**, **the leg** or **the shank** and **the foot**. **The femur** is the single bone of the thigh. It is the largest, longest and strongest bone in the body. The head of femur articulates with **the acetabulum** of the pelvis to form the hip joint. Two parallel bones, **the tibia** and **fibula**, form the skeleton of the shank. A thick, triangular **patella** forms the knee cap, which protects the knee joint anteriorly and improves the